

FUNCTION GENERATOR

Using ICL8038

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A function generator produces square, sine, and triangular waveforms across frequencies up to 100kHz. It plays a vital role in testing, repairing, and developing electronic equipment, serving as a signal source for amplifier testing and frequency response analysis. At its core, an oscillator generates sine, square, or triangular waves, which are transformed into other waveforms through wave-shaping circuits. The breadboard prototype demonstrates this functionality, with the sine waveform output captured on an oscilloscope, as shown in Fig. 1.

Circuit and working

Fig. 2 shows the circuit diagram of the function generator using

the ICL8038, which produces sine, square, and triangular waveforms simultaneously with minimal external components. Capacitor C1 is charged and discharged by two current sources.

The ICL8038 incorporates two built-in constant current sources: Current Source 1 operates continuously, while Current Source 2 toggles via a flip-flop. Initially, Current Source 1 charges capacitor C1, causing a linear voltage increase. Upon reaching two-thirds of the supply voltage, comparator 1 triggers the flip-flop, engaging Current Source 2 to discharge the capacitor at twice the rate of Current Source 1. The discharge continues until the voltage drops to one-third of the supply voltage. At this point,

the flip-flop resets, restarting the cycle.

When the current sources are set to I and 2I, the voltage across the capacitor forms a triangular wave, and the flip-flop output is a square wave with a 50% duty cycle. These waveforms are fed to a buffer stage and made available at pin 3 and pin 9, respectively.

The current source levels can be adjusted using two external resistors connected between pin 4 and pin 5 of the IC and the supply. The output becomes asymmetrical if the current sources are set to values other than I and 2I. The triangular wave can then be modified to a sawtooth wave, and the square wave's duty cycle adjusted from 1% to 99%. A nonlinear network

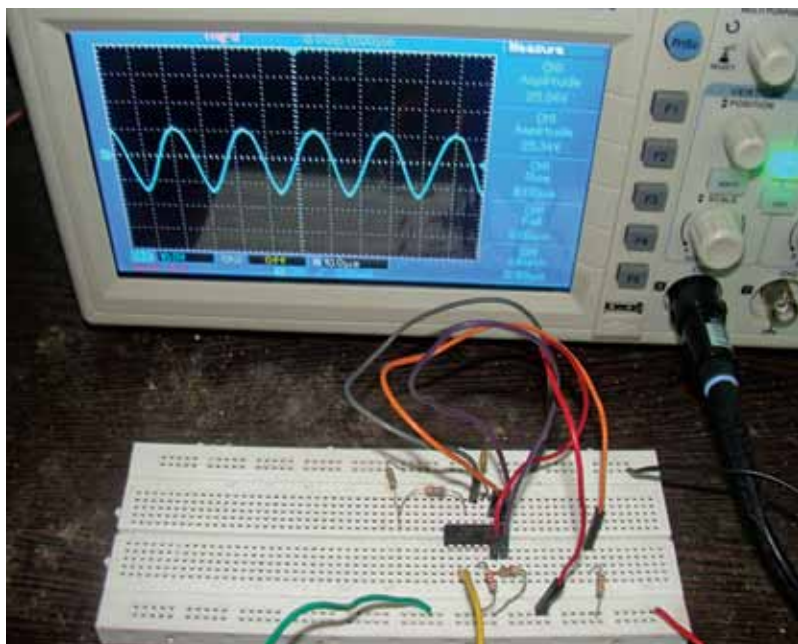


Fig. 1: Author's prototype

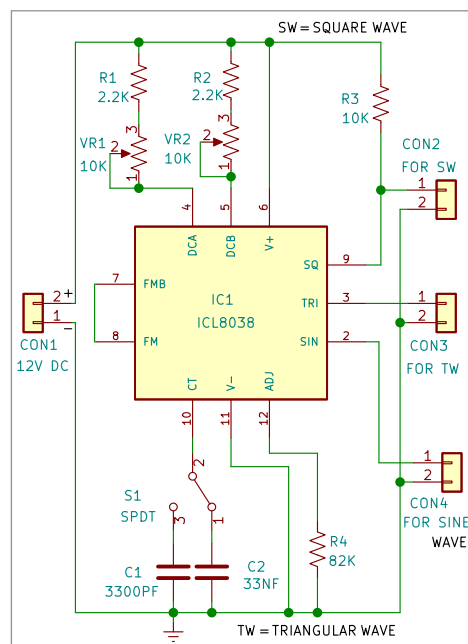


Fig. 2: Circuit diagram